

Biomolecules

NCERT Class XII Chemistry · Chapter 10 · Biomolecules

Biomolecules — Complete Notes

7 marks in the board paper (CBSE 2026-27 curriculum). This chapter is a tour of the molecules that living things are built from: sugars, proteins, enzymes, vitamins, DNA and RNA, hormones. It reads like biology, but the exam asks it as chemistry — a structure, a reagent, a reason. Every new word is tagged [exposure] and explained from scratch. Glucose is the heart of the chapter; get it and half the marks follow.

What a biomolecule is — *three definitions to hold on to before anything else*

[exposure] Biochemistry is the branch of science that studies the chemistry going on inside living things — what they are made of and which reactions run in them. It exists because a living body is just atoms and molecules following chemical rules; to understand digestion or heredity you have to understand the molecules. This chapter is a slice of it.

[exposure] A biomolecule is any complex organic molecule that builds up a living organism and is needed for its growth and upkeep. The four big families are **carbohydrates, proteins, lipids and nucleic acids**. This chapter covers three of them in full (carbohydrates, proteins, nucleic acids), then enzymes, vitamins and hormones.

[exposure] A polymer is a very long molecule made by joining many small identical or similar units, the **monomers**, end to end. Starch is a polymer of glucose; a protein is a polymer of amino acids; DNA is a polymer of nucleotides. You will see the same pattern three times, so keep it in mind: *small unit* → *joined by one specific bond* → *long chain*.

Examiner asks: nothing directly from this section, but it names the three monomer–polymer pairs the whole chapter is built on.

10.1 Carbohydrates — what counts as one — *the formula is a trap; the real definition is about groups*

[exposure] Carbohydrates are a huge group of naturally occurring organic compounds, made mostly by plants. Cane sugar, glucose and starch are all carbohydrates. They matter because they are the main fuel and the main structural material of living things. The name means "hydrate of carbon", because most of them fit the general formula $C_x(H_2O)_y$ — carbon with water attached.

Glucose $C_6H_{12}O_6$ fits it as $C_6(H_2O)_6$. But the formula cannot be the definition, in both directions:

- Acetic acid CH_3COOH fits it as $C_2(H_2O)_2$ and is **not** a carbohydrate.
- Rhamnose $C_6H_{12}O_5$ is a carbohydrate and does **not** fit it.

So the real definition is by structure: a carbohydrate is an **optically active polyhydroxy aldehyde or ketone**, or a compound that gives such units on hydrolysis. Three words to unpack:

[exposure] Optically active means the compound rotates the plane of plane-polarised light. A beam of light that vibrates in only one plane is plane-polarised; passing it through some compounds turns that plane to the right or left. It happens when a molecule has a carbon with four different groups on it (you met this as chirality in Haloalkanes). Why it matters here: it is the property that lets chemists tell sugars apart and track their structure.

[exposure] Polyhydroxy means more than one –OH group. An **aldehyde** is the group –CHO and a **ketone** is $>C=O$ inside a chain (the next chapter treats both properly). So a carbohydrate is a chain carrying several –OH groups and one –CHO or $>C=O$.

[exposure] Hydrolysis means breaking a bond by adding water, usually with a little acid or an enzyme to speed it up. The big carbohydrates are made by joining small ones and losing water, so hydrolysis runs it backwards and gives the small units back. Example: sucrose plus water, with a little acid, gives glucose plus fructose (below).

Carbohydrates that taste sweet are called **sugars**: the table sugar at home is **sucrose**; the sugar in milk is **lactose**. Carbohydrates are also called **saccharides**, from the Greek word for sugar.

Trap: " $C_x(H_{2O})_y$ means carbohydrate" is false in both directions. Cite acetic acid (fits, not a carbohydrate) and rhamnose (does not fit, is one). Then give the structural definition.

Classifying carbohydrates — *by what hydrolysis gives, and by whether they reduce Tollens' and Fehling's*

1. By behaviour on hydrolysis

[exposure] A **monosaccharide** is a carbohydrate that cannot be hydrolysed any further into simpler polyhydroxy aldehyde or ketone units. It is the single building block. About 20 are known in nature; glucose, fructose and ribose are the ones you need.

[exposure] An **oligosaccharide** gives 2 to 10 monosaccharide units on hydrolysis ("oligo" = few). They are named by count: **di-**, **tri-**, **tetra-saccharide** for 2, 3, 4 units. The common ones are disaccharides. Example: **sucrose** hydrolyses to one glucose plus one fructose (two different units); **maltose** hydrolyses to two glucose (two same units).

[exposure] A **polysaccharide** gives a large number of monosaccharide units on hydrolysis. Starch, cellulose, glycogen and gums are the standard examples. They are not sweet, so they are also called **non-sugars**.

2. By reducing action

[exposure] **Tollens' reagent** and **Fehling's solution** are two standard lab tests for an aldehyde: an aldehyde reduces them (and is itself oxidised), and you see a silver mirror or a red-brown precipitate. A carbohydrate that reduces either is a **reducing sugar**; one that does not is **non-reducing**. The test works only when a *free* aldehyde or ketone group is available to react.

- **All monosaccharides** are reducing sugars, aldose or ketose, because the group is free.
- A **disaccharide** is reducing if at least one of its two aldehyde/ketone groups is still free: **maltose**, **lactose**.
- It is **non-reducing** if both groups are tied up in the bond joining the two units: **sucrose**.

Examiner asks: "name the disaccharide that gives two glucose on hydrolysis" (2020 — maltose); "products of hydrolysis of sucrose" (2019 — glucose plus fructose); "which is a disaccharide, which is a monosaccharide" from a list of starch, maltose, glucose, fructose, cellulose (2015 — maltose; fructose); "why sucrose is non-reducing" (2010 — both reducing groups are used up in the bond).

Monosaccharides — the naming grid — *carbon count gives the stem; aldehyde or ketone gives the prefix*

[exposure] An **aldose** is a monosaccharide that carries an aldehyde group; a **ketose** carries a ketone group. The carbon count goes into the name too, so a six-carbon aldose is an **aldohexose** and a six-carbon ketose is a **ketohexose**. This two-part name tells you in one word what the molecule is built from.

Carbons	General term	Aldose	Ketose
3	triose	aldotriose	ketotriose
4	tetrose	aldotetrose	ketotetrose
5	pentose	aldopentose	ketopentose
6	hexose	aldohexose	ketohexose
7	heptose	aldoheptose	ketoheptose

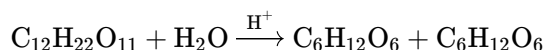
Glucose is an aldohexose. Fructose is a ketohexose. Ribose, which turns up in RNA, is an aldopentose.

Glucose — where it comes from — *two preparations, both ending in glucose*

Glucose occurs both free (sweet fruit, honey, ripe grapes) and combined (inside sucrose, starch, cellulose). It is made in the lab or industry in two ways.

From sucrose (cane sugar)

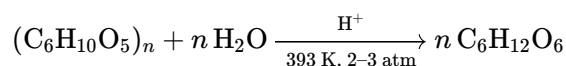
Boil sucrose with dilute HCl or H₂SO₄ in alcoholic solution. It hydrolyses to **equal amounts of glucose and fructose**. Both have the same molecular formula C₆H₁₂O₆.



Glucose and fructose are two different compounds with the same formula.

From starch (the industrial route)

Boil starch (or cellulose) with dilute H₂SO₄ at 393 K under 2–3 atm pressure. Every glucose unit in the chain is released.



Examiner asks: the conditions 393 K and 2–3 atm, and the fact that sucrose gives glucose plus fructose in equal amounts.

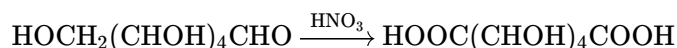
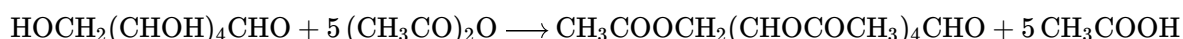
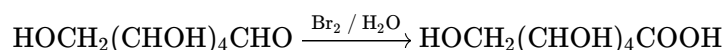
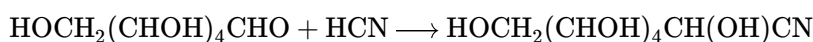
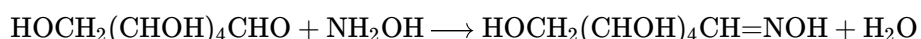
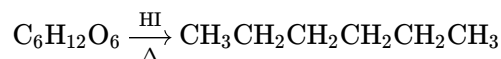
Glucose — proving its open-chain structure — *six reagents, six facts; this table is the single most repeated question in the chapter*

[exposure] The **open-chain structure** of glucose is the straight-chain form: six carbons in a row, an aldehyde group $-\text{CHO}$ at one end, a $-\text{CH}_2\text{OH}$ at the other, and four $-\text{CHOH}$ groups in between. Chemists did not see it; they inferred it by reacting glucose with chosen reagents and reading what came out. Each reaction proves one feature, and each one is a standalone exam question.



Reagent	What happens	What it proves
Molecular formula	$\text{C}_6\text{H}_{12}\text{O}_6$	six carbons
HI , prolonged heating	gives <i>n</i> -hexane $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$	all six carbons are in a straight chain
NH_2OH (hydroxylamine)	gives an oxime , $\text{CH}=\text{NOH}$	a carbonyl group is present
HCN	gives a cyanohydrin	a carbonyl group is present
Br_2 water (mild oxidant)	gives gluconic acid , $-\text{CHO} \rightarrow -\text{COOH}$	the carbonyl is an aldehyde , not a ketone
$(\text{CH}_3\text{CO})_2\text{O}$ (acetic anhydride)	gives glucose pentaacetate	five $-\text{OH}$ groups , each on a different carbon
HNO_3 (strong oxidant)	gives saccharic acid , a dicarboxylic acid	a primary alcohol $-\text{CH}_2\text{OH}$ is present

The reactions, written out:



[exposure] An **oxime** is what an aldehyde or ketone becomes after it condenses with hydroxylamine NH_2OH : the $\text{C}=\text{O}$ turns into $\text{C}=\text{N}-\text{OH}$ and one water leaves. It exists as a classic way to test for a carbonyl group. Example: glucose gives glucose oxime, $\text{CH}=\text{NOH}$ at the top of the chain.

[exposure] A **cyanohydrin** forms when HCN adds across a carbonyl: the carbon gains both a $-\text{CN}$ and an $-\text{OH}$. The CN^- ion is a nucleophile (an electron-rich attacker, as in Haloalkanes), it attacks the electron-

poor carbonyl carbon, and the oxygen then picks up H^+ from the solution. Same job as the oxime: it confirms a carbonyl.

[exposure] Acetylation means attaching an acetyl group CH_3CO- to an $-OH$, turning each one into an ester. Acetic anhydride does it. Glucose takes *five* acetyls, so it has five $-OH$ groups. Because the product is a stable compound, no two of those $-OH$ groups sit on the same carbon.

Trap: do not mix up which reagent proves which feature. Carbonyl is proved by *oxime or cyanohydrin* (they work on aldehydes *and* ketones). It is **bromine water** that narrows it to an aldehyde. And HNO_3 gives a *dicarboxylic acid* (both ends become $-COOH$), which is why NCERT concludes a **primary alcohol** is present.

Examiner asks: this table is the most repeated question in the chapter. Forms it takes: "write the product of glucose with HI" (2014); "with conc. HNO_3 " (2018); "show the open chain has a straight chain, five $-OH$, an aldehyde" (2019); "glucose with HCN, Br_2 , acetic anhydride" (2019); "heating glucose with hydroxylamine — which group is confirmed" (2023, answer: carbonyl). See the PYQ section.

Glucose — D, L and the (+) sign — *two labels that students merge into one; they mean different things*

[exposure] The **Fischer projection** is a way of drawing a 3-D molecule flat. The carbon chain runs vertically with the most oxidised carbon ($-CHO$) at the top. Each carbon sits where a horizontal line meets the vertical one. Horizontal bonds point *toward* you; vertical bonds point *away*. It exists because chemists needed one fixed convention to compare the 3-D arrangement of many sugars on paper. Fischer used it to fix the positions of glucose's $-OH$ groups after studying many of its properties.

In glucose's Fischer projection, going down from the aldehyde carbon, the $-OH$ groups sit **right, left, right, right** (on carbons 2, 3, 4, 5).

[exposure] An **asymmetric (chiral) carbon** is one attached to four different atoms or groups. A molecule with one comes in two mirror-image forms called **enantiomers**, and it is optically active. Glyceraldehyde ($HOCH_2-CH(OH)-CHO$) has exactly one, so it exists as two enantiomers. This is why it is chosen as the reference for the D/L system.

[exposure] **D and L** label the *spatial arrangement* of a compound relative to glyceraldehyde. Write the structure with the most oxidised carbon on top. Look at the **lowest asymmetric carbon** (the one furthest from the $-CHO$). If its $-OH$ is on the **right**, the compound is **D**; on the **left**, **L**. Glucose's lowest asymmetric carbon (C_5) has $-OH$ on the right, so glucose is **D-glucose**. Only that one carbon is used for the label.

[exposure] The **(+) or (-)** sign is a separate fact: it says which way the compound rotates plane-polarised light. (+) = dextrorotatory = to the right; (-) = laevorotatory = to the left. It has to be measured; you cannot read it off a drawing. Glucose rotates right, so it is **D-(+)-glucose**. The older lowercase d and l mean the same as (+) and (-).

Trap: D has no relation to optical rotation. D tells you the arrangement; (+) tells you the rotation direction. D-fructose is D but laevorotatory, written D-(-)-fructose. That one pair is the standard example to quote.

Quick relatives of glucose that follow from the same drawing: if the top $-CHO$ becomes $-COOH$ it is **gluconic acid**; if both ends are $-COOH$ it is **saccharic acid**.

Glucose — the ring form — *three facts the straight chain cannot explain; one idea that explains all of them*

The open-chain structure explains almost everything above, but three observations break it:

1. Glucose has an aldehyde group, yet it **does not give Schiff's test** (the pink-colour test for aldehydes) and it **does not form the hydrogensulphite addition product** with NaHSO_3 , which aldehydes normally do.
2. The **pentaacetate of glucose does not react with hydroxylamine**, though a free aldehyde would. So in glucose the $-\text{CHO}$ is not free.
3. Glucose forms **two crystalline forms**: the **α -form** (melting point 419 K, crystallised from concentrated solution at 303 K) and the **β -form** (melting point 423 K, crystallised from hot saturated solution at 371 K).

[exposure] A **hemiacetal** is what forms when an $-\text{OH}$ group adds to an aldehyde: the old carbonyl carbon ends up holding an $-\text{H}$, an $-\text{OH}$ and an $-\text{OR}$. When the $-\text{OH}$ and the aldehyde are on the *same molecule*, the addition closes a ring. This is the fix for all three facts: in glucose the $-\text{OH}$ on **carbon 5** adds to the carbon-1 aldehyde, and a **six-membered ring** (five carbons and one oxygen) closes. The $-\text{CHO}$ is now used up inside the ring, which is why Schiff's test, NaHSO_3 and hydroxylamine all fail.

The two crystal forms then have an easy explanation: closing the ring turns carbon 1 into a new asymmetric carbon, and the new $-\text{OH}$ on it can sit on either side. The two cyclic forms stay in **equilibrium with the open chain** in solution.

[exposure] The **anomeric carbon** is carbon 1, the former aldehyde carbon, now the carbon that holds the new $-\text{OH}$. The two ring forms differ *only* in the position of that one $-\text{OH}$: **on the right** in the Fischer drawing = **α -D-(+)-glucose**; **on the left** = **β -D-(+)-glucose**. Two isomers that differ only at the anomeric carbon are called **anomers**.

[exposure] A **pyranose** is any sugar ring with six atoms (five carbons and one oxygen), named after **pyran**, the ring compound with five carbons and one oxygen. So the glucose rings are **α -D-glucopyranose** and **β -D-glucopyranose**. The **Haworth structure** is the standard flat drawing of this ring, with the ring edge-on, and each group pointing either up or down.

Reading a Haworth drawing off the Fischer one

The rule that does every sugar: **on the right in Fischer $\rightarrow$ down in Haworth; on the left $\rightarrow$ up**. The terminal CH_2OH on carbon 5 points up in every D-sugar.

Carbon	α -D-glucopyranose	β -D-glucopyranose
C1 (anomeric)	$-\text{OH}$ down	$-\text{OH}$ up
C2	$-\text{OH}$ down	$-\text{OH}$ down
C3	$-\text{OH}$ up	$-\text{OH}$ up
C4	$-\text{OH}$ down	$-\text{OH}$ down
C5	$-\text{CH}_2\text{OH}$ up	$-\text{CH}_2\text{OH}$ up

The only line that changes between α and β is C1. That one change is the whole difference between the anomers.

Examiner asks: "enumerate the reactions of glucose that its open structure cannot explain" (2014); "why does glucose not give the 2,4-DNP test" (2019); "why the pentaacetate of glucose does not react with hydroxylamine" (2023); "define anomers" (2014). All four answers are the same idea — the --CHO is not free because it is inside the ring.

Fructose — *same formula as glucose, a ketone instead of an aldehyde, and a five-membered ring*

Fructose is the ketohexose that you get along with glucose from hydrolysing sucrose. It is found in fruit, honey and vegetables, and it is used as a sweetener. Its molecular formula is the same as glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, but the reactions show a **ketone group at carbon 2**, on the same six-carbon straight chain. It is a D-sugar (the lowest asymmetric carbon has --OH on the right) and it rotates light to the left, so its name is **D-(–)-fructose**.

[exposure] A **furanose** is a sugar ring with **five** atoms (four carbons and one oxygen), named after **furan**, the five-membered ring with four carbons and one oxygen. Fructose closes its ring when the --OH on **carbon 5** adds to the **carbon-2 ketone**. That ring is five-membered, unlike glucose's six, which is why the name is **fructofuranose**, not "-pyranose". Again there are two anomers, α and β , differing only at the new anomeric carbon (C2): **α -D-(–)-fructofuranose** and **β -D-(–)-fructofuranose**.

	Glucose	Fructose
Type	aldohexose	ketohexose
Ring formed by	$\text{C}_5 \text{ --OH} + \text{C}_1 \text{ aldehyde}$	$\text{C}_5 \text{ --OH} + \text{C}_2 \text{ ketone}$
Ring size	6 (pyranose)	5 (furanose)
Anomeric carbon	C1	C2
Rotation	(+) dextro	(–) laevo

Examiner asks: ring size and the names pyranose / furanose; the (–) sign on fructose.

Disaccharides — *two sugars joined through oxygen — sucrose, maltose, lactose, and which of them reduce*

[exposure] A **glycosidic linkage** is the C--O--C bridge that joins two monosaccharides. It forms when an --OH on one sugar and an --OH on the other lose one water between them, leaving a single oxygen as the link. NCERT calls this an "oxide linkage". This is the one bond that holds every disaccharide and polysaccharide together, and it is the carbohydrate equivalent of the peptide bond you will meet in proteins. A disaccharide hydrolysed with dilute acid or an enzyme gives back two monosaccharide molecules.

Disaccharide	Made of	Linkage	Reducing?
Sucrose	α -D-glucose + β -D-fructose	C1 of glucose – C2 of fructose	No
Maltose	two α -D-glucose	C1 of one – C4 of the other	Yes
Lactose	β -D-galactose + β -D-glucose	C1 of galactose – C4 of glucose	Yes

Sucrose

The common table sugar. On hydrolysis it gives an **equimolar mixture** (equal numbers of molecules) of D-(+)-glucose and D-(−)-fructose. In sucrose the link is made *through the aldehyde carbon of glucose (C1) and the ketone carbon of fructose (C2)*. Both reducing groups are used up in that one bond, so neither is free, and sucrose is a **non-reducing sugar**: it does not reduce Tollens' or Fehling's.

[exposure] Invert sugar is the glucose–fructose mixture you get by hydrolysing sucrose. Sucrose itself is dextrorotatory (+66.5°). But the hydrolysis gives glucose (+52.5°) and fructose (−92.4°), and the fructose rotates further left than glucose rotates right, so the **mixture is laevorotatory**. The sign of the rotation *inverts* from (+) to (−) during hydrolysis, which is where the name comes from.

Maltose

Two α-D-glucose units joined C1 to C4. The second glucose still has its C1 hemiacetal, which in solution opens to give a free aldehyde, so maltose **reduces**.

Lactose (milk sugar)

β-D-galactose plus β-D-glucose, joined C1 of galactose to C4 of glucose. The glucose unit still has a free C1, so lactose **reduces**. Galactose differs from glucose only at C4, where the −OH is on the other side. On hydrolysis, lactose gives β-D-glucose and β-D-galactose.

Trap: the reason for "sucrose is non-reducing" must say *both reducing groups, the aldehyde of glucose and the ketone of fructose, are involved in the glycosidic bond*. Saying "it has no aldehyde group" scores nothing; it does, until it is used in the bond.

Examiner asks: products of hydrolysis of lactose (2019); "why is sucrose non-reducing" (2010); name the linkage between two monosaccharides (2018, 2020 — glycosidic); "invert sugar" (see the PYQ section).

Polysaccharides — starch, cellulose, glycogen — the same monomer, three different jobs

Polysaccharides are thousands of monosaccharide units joined by glycosidic linkages. They do two jobs: **food storage** and **structure**.

Starch — the plant's stored food

The main storage polysaccharide of plants and our main dietary carbohydrate (cereals, roots, tubers). It is a polymer of α-glucose and has *two* components:

	Amylose	Amylopectin
Share of starch	15–20%	80–85%
Shape	long, unbranched chain, 200–1000 α-D-(+)-glucose units	branched chain of α-D-glucose
Linkage	C1–C4 glycosidic	C1–C4 along the chain; C1–C6 at the branches
In water	soluble	insoluble

Cellulose — the plant's structure

A straight-chain polymer of **β -D-glucose**, joined C1 to C4. It is the most abundant organic substance in the plant kingdom and the main constituent of plant cell walls. We use it as wood and as cotton.

Glycogen — the animal's stored food

The carbohydrate stored in the animal body, also called **animal starch**. Its structure is like amylopectin but **more highly branched**. It is found in the liver, muscles and brain; when the body needs glucose, enzymes break glycogen down. It is also found in yeast and fungi.

Trap: starch and cellulose are *both* glucose polymers. The structural difference they want is one line: **starch is α -D-glucose, cellulose is β -D-glucose**; starch has an unbranched *and* a branched part, cellulose is purely linear.

Why carbohydrates matter

They are the major part of our food; honey has long been used as an instant energy source by "vaidis" in the Ayurvedic system; starch and glycogen are the storage molecules; cellulose builds plant cell walls (NCERT adds bacteria) and gives us wood and cotton fibre. D-ribose and 2-deoxy-D-ribose are the two aldopentoses in nucleic acids, and carbohydrates occur in the body combined with proteins and lipids.

Examiner asks: a carbohydrate found in liver, muscle and brain (2019 — glycogen); the structural difference between starch and cellulose (2016, 2023); the two components of starch (2013); amylose versus amylopectin (2019).

10.2 Amino acids — the monomer of every protein — what one is, how they are classified, and why they behave like salts

[exposure] Proteins are the most abundant biomolecules in living systems; they form the basis of structure and function and are needed for growth and upkeep. Chief sources: milk, cheese, pulses, peanuts, fish, meat. The word comes from the Greek for "of primary importance". Every protein is a polymer of **α -amino acids**.

[exposure] An amino acid is a compound that carries both an amino group $-\text{NH}_2$ (basic) and a carboxyl group $-\text{COOH}$ (acidic). In an **α -amino acid** both sit on the same carbon, the α -carbon, along with a hydrogen and a side chain R: $\text{H}_2\text{N}-\text{CHR}-\text{COOH}$. Only α -amino acids come out of the hydrolysis of proteins. Different R groups make different amino acids. Example: R = H gives **glycine**; R = CH_3 gives **alanine**.

About 20 amino acids make up all plant and animal proteins. Each has a **trivial name** that records a property or the source — glycine is named from the Greek for "sweet" (it tastes sweet) and tyrosine from the Greek for "cheese" (it was first isolated from cheese). Each has a three-letter code and a one-letter code: glycine Gly / G, alanine Ala / A.

Classification 1: acidic, basic, neutral

Count the groups. **Equal** numbers of $-\text{NH}_2$ and $-\text{COOH} \rightarrow$ **neutral** (glycine, alanine). **More** $-\text{COOH} \rightarrow$ **acidic** (extra $-\text{COOH}$ in the side chain: aspartic acid, glutamic acid). **More** $-\text{NH}_2 \rightarrow$ **basic** (extra $-\text{NH}_2$ in the side chain: lysine, arginine).

Classification 2: essential and non-essential

[exposure] An **essential amino acid** is one the body cannot make, so it has to come from food. A **non-essential** one the body can make itself. Of the 20, **10 are essential** and 10 are not. Essential: valine, leucine, isoleucine, arginine, lysine, threonine, methionine, phenylalanine, tryptophan, histidine. Non-essential: glycine, alanine, glutamic acid, aspartic acid, glutamine, asparagine, serine, cysteine, tyrosine, proline.

Trap: for a two-example answer, safe picks are **valine and leucine** (essential) and **alanine and glutamic acid** (non-essential). Arginine and lysine are on *both* the "basic" list and the "essential" list, so do not use them as a clean example of either.

Properties

Amino acids are colourless crystalline solids, soluble in water, with high melting points. They behave like **salts** rather than like a simple amine or acid. The reason is the two groups in one molecule:

[exposure] A **zwitterion** (zwitter ion) is a molecule that carries a positive and a negative charge at the same time and is neutral overall. In water, the --COOH gives up H^+ to become --COO^- while the --NH_2 picks it up to become --NH_3^+ . Both charges sit on the same molecule: $\text{H}_3\text{N}^+ \text{--CHR--COO}^-$. That ionic character is why amino acids behave like salts, dissolve in water and melt at high temperature.

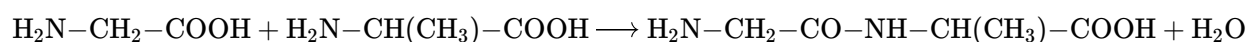
[exposure] **Amphoteric** means able to react with both acids and bases. In zwitterion form, an amino acid has an acidic group *and* a basic group, so it reacts with either. That is the whole answer to "why are amino acids amphoteric".

Optical activity. Every α -amino acid except **glycine** is optically active, because the α -carbon is asymmetric. Glycine's α -carbon carries two hydrogens, so it is not. Amino acids exist as D and L forms; the natural ones are **L**, written with the --NH_2 on the **left**. (For sugars you look at the --OH ; for amino acids you look at the --NH_2 .)

Examiner asks: define an amino acid (2014); what is an essential amino acid with examples (2014, 2018, 2025); why amino acids are amphoteric (2018, 2023, 2025); why they behave like salts (2023).

Peptides and the peptide bond — *how amino acids join; when a chain stops being a peptide and becomes a protein*

[exposure] A **peptide bond** (peptide linkage) is the amide bond --CO--NH-- that joins two amino acids. It forms when the --COOH of one amino acid reacts with the --NH_2 of the next and a molecule of water leaves. It exists because that is the only way nature strings amino acids into a chain; it is the protein version of the glycosidic linkage.



That is glycine's carboxyl joining alanine's amino group, giving the dipeptide **glycylalanine**.

Name by count: 2 amino acids $\rightarrow$ **dipeptide**; 3 $\rightarrow$ **tripeptide**; 4, 5, 6 $\rightarrow$ **tetra-, penta-, hexapeptide**; **more than 10** $\rightarrow$ **polypeptide**. Each amino acid inside the chain is called a **residue**, and the chain loses one water per bond: **n amino acids make n – 1 peptide bonds**.

A polypeptide with **more than 100 amino acid residues and a molecular mass above 10,000 u** is called a protein. The line is not sharp: a shorter chain still counts as a protein if it folds into a well-defined

protein shape, and the standard example is **insulin, with only 51 amino acids**.

Trap: glycosidic versus peptide linkage is a standing 2019 question. One-liner: **glycosidic** = C–O–C between two monosaccharides (carbohydrates); **peptide** = –CO–NH– amide between two amino acids, with loss of water (proteins).

Protein structure and shape — *fibrous or globular; then four levels, each one built on the last*

By overall shape

[exposure] **Fibrous proteins** are thread-like: many polypeptide chains lie side by side, held together by **hydrogen bonds and disulphide bonds** (S–S bridges between sulphur-containing amino acids). They are **insoluble in water**. Examples: **keratin** (hair, wool, silk) and **myosin** (muscle).

[exposure] **Globular proteins** are chains coiled up into a ball. They are generally **soluble in water**, which is why they can travel in blood plasma. Examples: **insulin and albumins**.

The four levels

1. **Primary** — the *sequence* of amino acids in each chain. Change even one amino acid in the sequence and you have a different protein. **Secondary** — the regular shape the backbone folds into, held by **hydrogen bonds** between the C=O of one peptide bond and the N–H of another. Two forms:
 - **α-helix**: one chain twists into a right-handed screw; each N–H bonds to the C=O one turn along. The hydrogen bonds are **within one chain** (intramolecular).
 - **β-pleated sheet**: several chains stretched out and laid side by side like folds in a curtain; the hydrogen bonds are **between different chains** (intermolecular).
2. **Tertiary** — the overall folding of the whole chain, giving the fibrous or globular shape. Held by hydrogen bonds, disulphide linkages, van der Waals forces and electrostatic attraction.
3. **Quaternary** — only in proteins with **two or more chains** (sub-units): the arrangement of the sub-units relative to each other. Haemoglobin is the example: four sub-units.

Trap: the α-helix versus β-sheet question is always about the *location* of the hydrogen bonds: **intramolecular (one chain) versus intermolecular (different chains)**. State both halves.

Examiner asks: an example each of fibrous and globular protein (2016); difference between α-helix and β-pleated sheet (2018, 2023); define primary structure (2024); two differences between globular and fibrous (2013).

Denaturation — *what boiling an egg does to a protein*

[exposure] A **native protein** is a protein as it occurs in a living system, with its own unique 3-D structure and its biological activity (an enzyme, for instance). **Denaturation** is what happens when a native protein is pushed out of that shape by a *physical* change (heat) or a *chemical* change (a change in pH). Hydrogen bonds break, globules unfold, helices uncoil, and the protein **loses its biological activity**. What stays: the **primary structure**, because it is held by peptide bonds, not hydrogen bonds. What is destroyed: the **secondary and tertiary** structure.

Two standard examples: **coagulation of egg white on boiling**, and **curdling of milk** (bacteria make lactic acid, the pH drops, the milk protein coagulates).

Examiner asks: define native and denatured protein with an example (2018, 2025); what changes in egg protein on boiling (2013 — it denatures, hydrogen bonds break, activity lost).

10.3 Enzymes — *biological catalysts — very specific, needed only in tiny amounts*

[exposure] Enzymes are biological catalysts: substances that speed up the chemical reactions of living things without being used up. They exist because the body has to run huge numbers of reactions under mild conditions (body temperature, neutral pH), and they only work fast enough with a catalyst. Almost every enzyme is a **globular protein**. Example: the enzyme **maltase** speeds the hydrolysis of maltose into glucose.

- **Specific** — one enzyme for one reaction and one substrate. The **substrate** is the molecule the enzyme acts on.
- **Needed only in small amounts** for the reaction to proceed.
- **Naming:** usually after the compound it works on (maltose → *maltase*), sometimes after the reaction (an enzyme catalysing oxidation of one substrate with simultaneous reduction of another is an **oxidoreductase**). The name **ends in "-ase"**.

How they work: like any catalyst, an enzyme lowers the **activation energy** (the energy barrier a reaction has to cross; you met it in Chemical Kinetics). NCERT's number: acid hydrolysis of sucrose needs 6.22 kJ mol^{-1} ; with the enzyme sucrase it needs only 2.15 kJ mol^{-1} .

Examiner asks: define enzymes (2014 — biological catalysts, almost all globular proteins).

10.4 Vitamins — *definition, the two classes, and the source–disease table that is a pure recall mark*

[exposure] Vitamins are organic compounds needed in the diet in *small* amounts to perform specific biological functions for normal growth and health. Their deficiency causes specific diseases. Most cannot be made in the human body (so food has to supply them); plants make almost all of them, so vitamins count as "essential food factors". They are named A, B, C, D, E, K, with sub-groups such as B₁, B₂, B₆, B₁₂.

Two classes, by solubility

- **Fat-soluble** — dissolve in fat and oil, not in water: **A, D, E, K**. They are stored in the liver and in adipose (fat-storing) tissue.
- **Water-soluble** — **B group and C**. They must be taken **regularly** because they are readily excreted in urine and cannot be stored — **except vitamin B₁₂**, which is stored in the liver.

Vitamin	Source	Deficiency disease
A (retinol)	fish liver oil, carrot, butter, milk	xerophthalmia (hardening of the cornea); night blindness
B ₁ (thiamine)	yeast, milk, green vegetables, cereals	beri-beri (loss of appetite, retarded growth)
B ₂ (riboflavin)	milk, egg white, liver, kidney	cheilosis (fissures at corners of mouth and lips), digestive disorders, burning skin
B ₆ (pyridoxine)	yeast, milk, egg yolk, cereals, gram	convulsions
B ₁₂	meat, fish, egg, curd	pernicious anaemia (red blood cells deficient in haemoglobin)
C (ascorbic acid)	citrus fruits, amla, green leafy vegetables	scurvy (bleeding gums)
D	sunlight, fish, egg yolk	rickets (bone deformity in children); osteomalacia (soft bones, joint pain in adults)
E	vegetable oils (wheat germ, sunflower)	increased fragility of red blood cells; muscular weakness
K	green leafy vegetables	increased blood-clotting time

Vitamin D is the one that the body can make itself, when skin is exposed to sunlight; every other vitamin has to come from the diet.

Trap: "why can vitamin C not be stored in the body" (2016) — **it is water-soluble and is excreted in urine**. And the 1-mark favourites: bleeding gums → C; pernicious anaemia → B₁₂; bone deformity in children → D; blood clotting → K.

Examiner asks: define vitamin and classify it by solubility; diseases for A, E, C, D, B₁₂; which vitamin the body can make.

10.5 Nucleic acids — what they are made of — *sugar + phosphate + a nitrogen base, then how those units chain together*

[exposure] Heredity is the passing of inherited characters from one generation to the next. It is carried by **chromosomes**, thread-like structures in the cell nucleus made of protein plus a second biomolecule: the **nucleic acids**. They come in two types, **DNA** (deoxyribonucleic acid) and **RNA** (ribonucleic acid). Because each is a long chain polymer of **nucleotides**, nucleic acids are also called **polynucleotides**.

Complete hydrolysis gives three things

1. A **pentose sugar** (five-carbon, in ring form): **β-D-2-deoxyribose** in DNA, **β-D-ribose** in RNA. They differ only at carbon 2: ribose has an –OH there; 2-deoxyribose has just –H (no oxygen).
2. **Phosphoric acid** — the same in both.
3. A **nitrogen base**.

[exposure] The **nitrogen bases** are nitrogen-containing ring compounds ("heterocyclic": a ring containing something other than carbon). Two families: **purines** (double ring): **adenine (A)** and **guanine (G)**; and **pyrimidines** (single ring): **cytosine (C)**, **thymine (T)**, **uracil (U)**. DNA has A, G, C and T;

RNA has A, G, C and U. Thymine is uracil with an extra $-\text{CH}_3$. So thymine is found only in DNA, and uracil only in RNA.

Nucleoside and nucleotide

[exposure] A **nucleoside** is a **base joined to the 1' carbon of the sugar**. The carbons of the sugar are written 1', 2', 3', ... (read "one prime") so they cannot be mixed up with the carbons in the base. A

nucleotide is a nucleoside with **phosphoric acid attached at the 5' position** of the sugar. So:

nucleoside = sugar + base; nucleotide = sugar + base + phosphate.

[exposure] A **phosphodiester linkage** is how nucleotides chain up. The phosphate on the **5' carbon** of one nucleotide bonds to the $-\text{OH}$ on the **3' carbon** of the next, so the phosphate group is joined to two sugars (an ester at each end, hence "di-ester"). Repeat and you get a polynucleotide whose backbone alternates sugar–phosphate–sugar–phosphate, with the bases hanging off the sides. One end of the chain has a free phosphate on the 5' carbon (the **5' end**); the other has a free $-\text{OH}$ on a 3' carbon (the **3' end**).

Trap: "base + sugar" is the nucleoside; adding the phosphate makes the nucleotide. Questions on nucleoside versus nucleotide (2016, 2019) are lost on one missing word: **phosphate**.

Examiner asks: the unit formed by attaching a base to the 1' position of a sugar (2020 — nucleoside); the type of linkage in nucleic acids (2016 — phosphodiester); nucleoside versus nucleotide (2016, 2019); define nucleotide (2025).

Nucleic acids — structure and functions — *primary, the DNA double helix, base pairing, three kinds of RNA, three jobs of DNA*

The **primary structure** of a nucleic acid is the *sequence of nucleotides* along the chain — which bases come in which order.

DNA secondary structure — the double helix. Watson and Crick showed that DNA is two polynucleotide strands wound around each other in a helix, running in **opposite directions** (one $5' \rightarrow 3'$, the other $3' \rightarrow 5'$), and held together by **hydrogen bonds between pairs of bases**. The pairing is fixed:

- **Adenine pairs with thymine** — 2 hydrogen bonds ($\text{A}=\text{T}$).
- **Guanine pairs with cytosine** — 3 hydrogen bonds ($\text{G}\equiv\text{C}$).

That is why the two strands are **complementary**, not identical: wherever one has A, the other has T, and G faces C. It also gives the double helix its stability.

RNA secondary structure. RNA is normally a **single-stranded** helix; it sometimes folds back on itself. There are three types, each with a different job: **messenger RNA (m-RNA)** carries the message from DNA; **ribosomal RNA (r-RNA)** sits in ribosomes and helps in protein synthesis; **transfer RNA (t-RNA)** brings amino acids for protein synthesis.

	DNA	RNA
Strands	double helix	single strand
Sugar	deoxyribose	ribose
Bases	A, G, C, T	A, G, C, U
Main job	heredity — carries genetic information to the next generation	protein synthesis

Biological functions of nucleic acids

1. **Heredity** — DNA is the chemical basis of heredity and the store of genetic information.
2. **Replication** — during cell division the DNA makes a copy of itself, and identical DNA goes to both daughter cells.
3. **Protein synthesis** — the various RNAs make the proteins, but the *message* for each protein is in the DNA, passed on through m-RNA.

Examiner asks: "two strands in DNA are not identical but complementary" (2019); structural and functional differences between DNA and RNA (2013, 2023); one function of nucleic acids (2013).

10.6 Hormones — *chemical messengers* — *three chemical types and what each does*

[exposure] Hormones are molecules that act as **intercellular messengers**: they carry a signal from one group of cells or tissue to another. They are made by **endocrine (ductless) glands** and poured directly into the blood, which carries them to receptors at the site of action. They exist so the body can coordinate itself over distance. Compare with vitamins: hormones are **made in the body** and are messengers; vitamins (except D) have to come from food and are needed in small amounts for growth.

Chemical type	Examples	Role
Steroids (four-ring skeleton)	oestrogens, androgens	sexual development
Polypeptides (chains of amino acids)	insulin, endorphins	insulin lowers blood glucose; endorphins reduce pain
Amino-acid derivatives	epinephrine, norepinephrine	mediate the stress response

What they do

- **Glucose balance** — **insulin** lowers blood glucose (it turns glucose into glycogen); **glucagon** raises it (glycogen back to glucose). The two together hold the level steady.
- **Stress response** — epinephrine and norepinephrine.
- **Growth and development** — growth hormones (height, muscle mass); sex hormones (testosterone, oestradiol) for secondary sexual characters.
- **Thyroid** — the thyroid gland makes **thyroxine**, an iodinated derivative of tyrosine, which controls metabolism. Too little gives hypothyroidism; too much gives hyperthyroidism. A diet low in iodine causes enlargement of the thyroid (**goitre**); iodised salt prevents it.
- **Adrenal cortex** — **glucocorticoids** control carbohydrate metabolism and inflammation; **mineralocorticoids** control the kidney's salt and water balance. If the adrenal cortex fails the result is **Addison's disease**.
- **Progesterone** — prepares the uterus for implantation of the fertilised egg.

Examiner asks: how vitamins and hormones differ in source and function (2013 — hormones made by endocrine glands and released into blood, they transfer information between cells; vitamins come from the diet, in minute quantity, for growth and upkeep).

Patterns, collected — *the conversions, counts and calls you will actually be asked to make*

1. Reagent → product for glucose

Worked example

Glucose is heated with HI for a long time. Name the product and say what it proves.

Straight-chain six-carbon alkane: $\text{C}_6\text{H}_{12}\text{O}_6 \xrightarrow[\Delta]{\text{HI}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3 = n\text{-hexane}$. All six carbons lie in a straight chain, with no branch.

- Glucose + Br_2 water: product and conclusion — answer only: gluconic acid; the carbonyl is an aldehyde
- Glucose + acetic anhydride: product and conclusion — answer only: glucose pentaacetate; five $-\text{OH}$ groups
- Glucose + conc. HNO_3 : product and conclusion — answer only: saccharic acid (dicarboxylic acid); a primary $-\text{OH}$ is present

2. "Why does glucose not give ... but the open chain says it should?"

Worked example

Why does glucose not give Schiff's test even though it has an aldehyde group?

In solution the $-\text{OH}$ on C5 adds to the C1 aldehyde (intramolecular hemiacetal formation) and a six-membered ring forms. The $-\text{CHO}$ is no longer free, so no test that needs a free $-\text{CHO}$ can succeed. The same answer covers NaHSO_3 , 2,4-DNP, and hydroxylamine on the pentaacetate.

- Why does glucose pentaacetate not react with hydroxylamine? — answer only: the ring locks the $-\text{CHO}$; no free $-\text{CHO}$

3. Reading D or L from a drawing

Worked example

A Fischer drawing of a sugar has $-\text{OH}$ on the *right* at carbons 2, 4, 5 and on the *left* at carbon 3. D or L?

1. Put the most oxidised carbon ($-\text{CHO}$) on top — it already is.
 2. Find the lowest asymmetric carbon: C5 (C6 is CH_2OH , not asymmetric).
 3. Its $-\text{OH}$ is on the right → **D**. The other carbons are ignored. (This is D-glucose.)
- The same sugar with $-\text{OH}$ on the left at C5 — answer only: L
 - Amino acid with $-\text{NH}_2$ on the left of the α -carbon — answer only: L-amino acid (the natural form)

4. Haworth drawing from the Fischer

Worked example

D-glucose, Fischer: $-\text{OH}$ right at C2, left at C3, right at C4, right at C5. Draw β -D-glucopyranose.

Right → down, left → up: C2 down, C3 up, C4 down. $-\text{CH}_2\text{OH}$ on C5 is up. For β the anomeric $-\text{OH}$ on C1 is **up** (for α it would be down).

- Where is the C1 $-\text{OH}$ in α -D-glucopyranose? — answer only: down

5. Sign flip on hydrolysing sucrose

Worked example

Show that hydrolysed sucrose is laevorotatory, using the rotations $+52.5^\circ$ (glucose) and -92.4° (fructose).

Hydrolysis gives glucose and fructose in *equal* molar amounts, so each contributes equally. Mixture rotation $\approx \frac{(+52.5) + (-92.4)}{2} = -19.95^\circ$ — negative. Sucrose started at $+66.5^\circ$, so the sign has *inverted*, hence "invert sugar". (The -19.95° is the average built from the two NCERT values; use it to justify the sign, not as a quoted constant.)

- Which monosaccharide, glucose or fructose, makes the mixture laevorotatory, and why? — answer only: fructose; its -92.4° outweighs glucose's $+52.5^\circ$

6. Is this disaccharide reducing?

Worked example

Maltose and sucrose: which reduces Fehling's?

Check whether either aldehyde/ketone carbon is free. Maltose: the second glucose's C1 is not in the bond (the bond is C1 of the first unit to C4 of the second), so it can open to a free aldehyde → reducing. Sucrose: the bond is C1 of glucose to C2 of fructose, which uses *both* reducing carbons → non-reducing.

- Lactose — reducing? — answer only: yes (C1 of the glucose unit is free)

7. Counting peptides

Worked example

How many peptide bonds and how many water molecules are lost when 5 amino acids join in a chain?

Each bond forms one H_2O and joining n units takes $n - 1$ bonds: $5 - 1 = 4$ peptide bonds and **4** water molecules (a pentapeptide).

- A tripeptide — how many peptide bonds? — answer only: 2
- Which structure level is lost on denaturation: 1° , 2° , 3° ? — answer only: 2° and 3° (1° stays)

8. Pairing the bases

Worked example

One DNA strand reads $5'-\text{A}-\text{T}-\text{G}-\text{C}-3'$. Write the complementary strand and count the hydrogen bonds.

$\text{A} \rightarrow \text{T}$, $\text{T} \rightarrow \text{A}$, $\text{G} \rightarrow \text{C}$, $\text{C} \rightarrow \text{G}$, giving $3'-\text{T}-\text{A}-\text{C}-\text{G}-5'$. Hydrogen bonds: A–T has 2, T–A has 2, G–C has 3, C–G has 3 → **10**.

- A strand $5'-\text{G}-\text{G}-\text{A}-3'$: complementary strand and hydrogen bonds — answer only: $3'-\text{C}-\text{C}-\text{T}-5'$; $3 + 3 + 2 = 8$

Past year questions · question types — ranked by how often they turn up

Two videos feed this section. [30+] is the "30+ PYQ" video for CBSE 2024-25 — Q1–Q32, questions from 2013 to 2023. [2025] is the short "2025 PYQ" video — Q1–Q6, from the 2025 board paper. Question numbers below are the video's own numbering (the year comes from the video too). Where the lecture video solved a past question but gave only the year, it is cited as "year · one-shot". The CBSE 2026-27 "25 PYQ" video for this chapter was not uploaded when these notes were built; once it is, this section can be refreshed from it. Counts are by question part, from the two videos' lists.

1 · Glucose: reagent → product, and what it proves — 1, 2 or 3 marks · about 10 parts

Recognise it: "glucose is treated with ...", "write reactions to show that glucose contains ...", or "why does glucose not give ... test".

1. Name the feature the reagent tests.
2. Write the one-line reaction: HI $\rightarrow$ *n*-hexane; Br₂ water $\rightarrow$ gluconic acid; (CH₃CO)₂O $\rightarrow$ pentaacetate; HNO₃ $\rightarrow$ saccharic acid; NH₂OH $\rightarrow$ oxime; HCN $\rightarrow$ cyanohydrin.
3. Finish with the conclusion sentence — "this shows ...". The conclusion is usually the mark.
4. If it says "why does glucose *not* give ...", the answer is always the ring: the —CHO is tied up in the hemiacetal, so it is not free.

Trap: HI needs *prolonged* heating to reach *n*-hexane. Br₂ water proves an **aldehyde** (it does not touch a ketone); oxime and cyanohydrin only prove a **carbonyl**. Mixing those two conclusions loses the mark.

2 · Differentiate / write the difference — 1 to 3 marks · about 9 parts

Recognise it: "differentiate between ...", "write one structural difference ...", "how are X and Y different in source and function".

1. Give each side one line, *on the same property*.
2. Use the pair you were given: starch/cellulose $\rightarrow$ α - versus β -glucose; amylose/amylopectin $\rightarrow$ unbranched versus branched, soluble versus insoluble; DNA/RNA $\rightarrow$ double versus single strand, deoxyribose versus ribose, T versus U; nucleoside/nucleotide $\rightarrow$ missing versus present phosphate; fibrous/globular $\rightarrow$ insoluble versus soluble; α -helix/ β -sheet $\rightarrow$ intra- versus intermolecular hydrogen bonding.
3. For 3 marks give three contrasts (monomer, shape, solubility, linkage, function).

Trap: two separate definitions score less than one contrast. For α -helix versus β -sheet the answer is *where the hydrogen bonds are* — and you must say both.

3 · Define the term — 1 or 2 marks · about 8 parts

Recognise it: "define ...", "what is meant by ...". The terms that have appeared: amino acid, anomers, enzymes, oligosaccharides, invert sugar, denatured protein, native protein, nucleotide, essential amino acid.

1. One-line definition built on the key word.
2. Add the example — it is often a separate mark. Denatured protein $\rightarrow$ boiled egg white; essential amino acid $\rightarrow$ valine, leucine; invert sugar $\rightarrow$ hydrolysed sucrose; oligosaccharide $\rightarrow$ sucrose, lactose.

Trap: an "define with an example" question (2018 Q20) gives nothing for a bare definition. And "invert sugar" must say the *sign of rotation changes from (+) to (–)*.

4 · Name / identify — 1 mark · about 8 parts

Recognise it: "name the ...", "which of the following is a disaccharide", "write the product of hydrolysis of ...".

1. Answer in one line, no working.
2. Learn the short list: two glucose $\rightarrow$ **maltose**; sucrose $\rightarrow$ glucose + fructose; **lactose** $\rightarrow$ β -D-glucose + β -D-galactose; linkage between monosaccharides $\rightarrow$ **glycosidic**; linkage in nucleic acids $\rightarrow$ **phosphodiester**; the two parts of starch $\rightarrow$ **amylose, amylopectin**.

Trap: the "which is a disaccharide" list always offers starch (polysaccharide), glucose and fructose (monosaccharides), and a decoy like amylose or cellulose. Only maltose, sucrose or lactose is the answer.

5 · Give a reason — 2 or 3 marks · about 6 parts

Recognise it: "why are amino acids amphoteric", "why do amino acids behave like salts", "what changes in egg protein on boiling", "the two DNA strands are complementary, not identical", "water-soluble vitamins must be taken regularly".

1. State the structural fact (acidic and basic group in one molecule; hydrogen bonds break; A pairs with T and G with C; water-soluble so excreted in urine).
2. Say how the fact causes the behaviour.

Trap: vitamin B₁₂ is the one water-soluble vitamin that is stored (liver). Put "except B₁₂" in the answer when the question is about storage.

6 · Vitamin recall — 1 or 3 marks · about 5 parts

Recognise it: "the vitamin whose deficiency causes ...", "classify vitamins and name the one for blood clotting", "the only vitamin made in the body".

1. Classify by solubility: fat-soluble A, D, E, K; water-soluble B group and C.
2. Match the disease from the table in the lecture section: bleeding gums → C; bone deformity → D; night blindness → A; muscular weakness → E; clotting → K.

Trap: blood coagulation is vitamin **K**, and it is also the vitamin whose lack *increases* clotting time. Vitamin D is the one made in the body (sunlight).

Past year questions · mark slots — *what each type is worth*

The 30+ video groups its questions by mark value: Q1–Q18 are 1-mark, Q19–Q23 are 2-mark, Q24–Q32 are 3-mark. Where it names a different value for one question, that is shown. Biomolecules does not appear as a 5-mark long answer in either video.

Question type	1 mark	2 marks	3 marks
Glucose reagent / evidence	2014 Q2, 2018 Q4 [30+]	2023 Q31 [30+]; 2025 Q3, Q5 [2025]	2019 Q25, 2019 Q30 [30+]
Differentiate	2016 Q9, 2016 Q11, 2013 Q15 [30+]	2013 Q21, 2018 Q23(b) [30+]	2013 Q26(a), 2019 Q29 [30+]; 2025 Q6(a) [2025]
Define	2014 Q6, Q13, Q17 [30+]	Q19, 2018 Q20 [30+]; 2025 Q2 [2025]	2014 Q27 [30+]; 2025 Q4 [2025]
Name / identify	2020 Q3, 2016 Q5, 2019 Q8, 2013 Q12, 2015 Q14, 2020 Q16 [30+]	2025 Q1 [2025]	2025 Q6(b) [2025] (inside a 3-mark question)
Give a reason	—	2014 Q22, 2018 Q23(a) [30+]	2013 Q26(b), 2019 Q28, 2023 Q32 [30+]
Vitamin recall	2015 Q1, 2013 Q7, 2013 Q10 [30+]	—	2015 Q24 [30+]; 2025 Q6(c) [2025]

The video gives 2018 Q4 and the other items in the 1-mark block as 1-mark questions; Q15 (vitamins versus hormones) sits in that block too, though its answer is two paragraphs long (inferred: it would carry more marks in a real paper). The 2025 video states Q1 as a 2-mark question, but does not state a mark value for Q2 (inferred: 2).

Past year questions · repeat offenders — *appeared more than once* — highest probability in the chapter

2019 Q25 · 2019 Q30 · 2025 Q3(b) · 2025 Q5(a) · 2021 · one-shot

Glucose with bromine water → gluconic acid, to prove an aldehyde. The single most repeated reaction in the chapter.

2019 Q25 · 2019 Q30 · 2025 Q5(b) · 2014 Q2 · 2021 · one-shot

Acetic anhydride → glucose pentaacetate (five –OH), and HI prolonged heating → *n*-hexane (straight chain).

2019 Q30 · 2025 Q3(a) · 2021 · one-shot

Glucose with HCN → cyanohydrin. The CN^- attacks the carbonyl carbon.

2014 Q22 · 2019 Q28(a) · 2023 Q32(a)

Why glucose does not behave like an aldehyde — no Schiff's test, no NaHSO_3 adduct, no 2,4-DNP, and the pentaacetate does not react with hydroxylamine. One answer: the ring has locked the –CHO.

2014 Q27 · 2018 Q20 · 2025 Q2 · 2025 Q4(c)

Essential amino acid — definition and examples (valine, leucine). Four appearances in the two videos.

2018 Q23(a) · 2025 Q2 · 2023 · one-shot

Why amino acids are amphoteric — an acidic –COOH and a basic –NH₂ in the same molecule.

2018 Q20 · 2025 Q4(a) · 2025 Q6(a) · 2013 Q26(b)

Native versus denatured protein; boiling an egg. Hydrogen bonds break, biological activity is lost, the primary structure survives.

2019 Q8 · 2025 Q1 · 2019 · one-shot

Hydrolysis products of lactose (β -D-glucose + β -D-galactose) and of sucrose (D-glucose + D-fructose).

2015 Q14 · 2020 Q3 · 2025 Q6(b) · 2015 · one-shot

Pick the disaccharide from a list (maltose, lactose; never starch, glucose, fructose, amylose).

2015 Q24 · 2025 Q6(c)

The vitamin for blood coagulation — K.

2016 Q11 · 2025 Q4(b) · 2019 · one-shot · 2020 · one-shot

Nucleoside versus nucleotide; the unit formed when a base attaches to the 1' position of a sugar. The phosphate is the whole difference.

2020 Q16 · 2019 Q29(b) · 2018 · one-shot · 2019 · one-shot

Glycosidic linkage, and glycosidic versus peptide linkage.

2016 Q9 · 2013 Q12 · 2019 Q29(a) · 2023 · one-shot

Starch versus cellulose, and amylose versus amylopectin.

2013 Q21 · 2023 · one-shot

DNA versus RNA (strands, sugar; and function — heredity versus protein synthesis).

2018 Q23(b) · 2023 · one-shot

α -helix versus β -pleated sheet.

2023 Q32(c) · 2016 · one-shot

Why water-soluble vitamins (and vitamin C) must be taken regularly — excreted in urine, not stored (B₁₂ excepted).

Past year questions · numerical types — *no calculations in this chapter, so these are the reaction and structure types — one worked model each, then cold practice*

A · Glucose with a reagent

Worked example · 2019 Q30 [30+] — glucose with HCN, bromine water and acetic anhydride

HCN: CN^- is a nucleophile; it attacks the carbonyl carbon and the oxygen takes H^+ :

$\text{HOCH}_2(\text{CHOH})_4\text{CHO} + \text{HCN} \longrightarrow \text{HOCH}_2(\text{CHOH})_4\text{CH}(\text{OH})\text{CN}$. Product: **glucose cyanohydrin**. It confirms a carbonyl group.

Br_2 **water**: mild oxidant, takes the aldehyde to an acid:

$\text{HOCH}_2(\text{CHOH})_4\text{CHO} \xrightarrow{\text{Br}_2 / \text{H}_2\text{O}} \text{HOCH}_2(\text{CHOH})_4\text{COOH}$. Product: **gluconic acid**. It confirms an aldehyde.

Acetic anhydride: acetylates every $-\text{OH}$: product **glucose pentaacetate**. Five acetyl groups attach, so there are five $-\text{OH}$ groups.

- 2014 Q2 [30+] — product of glucose with HI — answer only: *n*-hexane on prolonged heating; six carbons in a straight chain
- 2018 Q4 [30+] — D-glucose with concentrated HNO_3 — answer only: saccharic acid (D-saccharic acid); aldehyde and primary $-\text{OH}$ both oxidised to $-\text{COOH}$
- 2023 Q31 [30+] — glucose heated with hydroxylamine, and the group confirmed — answer only: glucose oxime, $\text{CH}=\text{NOH}$; a carbonyl group
- 2025 Q5 [2025] — how to show the aldehyde group and the five $-\text{OH}$ groups — answer only: bromine water gives gluconic acid; acetic anhydride gives the pentaacetate

B · Why glucose does not behave like an aldehyde

Worked example · 2019 Q28(a) [30+] — why glucose does not give the 2,4-DNP test

The 2,4-DNP test needs a free $-\text{CHO}$ or $>\text{C}=\text{O}$. In solution the C5 $-\text{OH}$ adds to the C1 aldehyde and a six-membered hemiacetal ring forms. The aldehyde is no longer free, so there is no reaction. (Same answer for Schiff's test and for NaHSO_3 .)

- 2023 Q32(a) [30+] — why the pentaacetate of glucose does not react with hydroxylamine — answer only: the $-\text{CHO}$ is part of the ring, not free

- 2014 Q22 [30+] — reactions the open chain cannot explain — answer only: no NaHSO₃ adduct; no Schiff's or 2,4-DNP test; pentaacetate does not react with NH₂OH; two crystalline forms (α, β)

C · Hydrolysis products

Worked example · 2025 Q1 [2025] — hydrolysis products of sucrose and of lactose

Both are disaccharides, so each gives two monosaccharides in equal numbers. Sucrose gives **D-(+)-glucose and D-(-)-fructose** (invert sugar). Lactose gives **β-D-galactose and β-D-glucose**.

- 2019 Q8 [30+] — hydrolysis product of lactose — answer only: β-D-glucose and β-D-galactose
- 2020 Q3 [30+] — disaccharide that gives two glucose — answer only: maltose
- 2025 Q6(b) [2025] — which of glucose, lactose, amylose, fructose is a disaccharide — answer only: lactose

D · Differentiate a pair

Worked example · 2013 Q21 [30+] — structural and functional difference between RNA and DNA

Structure: DNA is a double helix (two strands); RNA is a single strand. DNA's sugar is deoxyribose; RNA's is ribose. (Bases: DNA has thymine, RNA has uracil.) **Function:** DNA carries genetic information from one generation to the next (heredity); RNA helps in protein synthesis.

- 2016 Q9 [30+] — structural difference between starch and cellulose — answer only: starch is α-D-glucose (linear and branched), cellulose is β-D-glucose (linear)
- 2018 Q23(b) [30+] — α-helix versus β-pleated sheet — answer only: intramolecular versus intermolecular hydrogen bonding
- 2019 Q29(a) [30+] — amylose versus amylopectin — answer only: unbranched and water-soluble (15–20%) versus branched and insoluble (80–85%)
- 2025 Q6(a) [2025] — native versus denatured protein — answer only: native keeps its 3-D structure and activity; denatured has lost its hydrogen bonding and activity after a physical or chemical change

E · Vitamin matching

Worked example · 2013 Q10 [30+] — deficiency diseases of vitamins A and E

Vitamin A → **night blindness** (and xerophthalmia). Vitamin E → **muscular weakness** (and fragile red blood cells).

- 2015 Q1 [30+] — vitamin whose lack causes bleeding gums — answer only: vitamin C
- 2013 Q7 [30+] — the vitamin the body can make, and its deficiency diseases — answer only: vitamin D; rickets in children, osteomalacia in adults
- 2015 Q24 [30+] / 2025 Q6(c) [2025] — vitamin for blood coagulation — answer only: vitamin K; vitamins are classified into fat-soluble (A, D, E, K) and water-soluble (B group, C)